

Growth Rate Hierarchy

Growth rate hierarchy ranks common complexity classes by how fast they grow as $n \rightarrow \infty$.

Natural Explanation

When input size gets very large, some algorithms scale gently (like logarithmic), while others explode (like exponential or factorial). The hierarchy is a way to compare this long-run behavior.

If $f(n)$ is lower in the hierarchy than $g(n)$, then eventually $f(n)$ is much smaller than $g(n)$, and algorithms with runtime $f(n)$ are typically more scalable.

Academic View

A standard way to state strict asymptotic separation is:

$$f(n) = o(g(n)) \iff \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0.$$

So when we write one class below another in the hierarchy, we usually mean the lower function is little-o of the higher one.

Common Hierarchy (Small to Large)

For sufficiently large n :

$$1 \prec \log \log n \prec \log n \prec (\log n)^k \prec n^\varepsilon \prec n \prec n \log n \prec n^2 \prec n^3 \prec n^k \prec c^n \prec n! \prec n^n$$

Typical assumptions: - $k > 1$ is a constant. - $0 < \varepsilon < 1$ is a constant. - $c > 1$ is a constant.

Read this left to right as increasing growth.

![[big-o-complexity.jpg]]

Useful Pairwise Facts

- $\log n = o(n^\varepsilon)$ for every constant $\varepsilon > 0$.
- $n^a = o(n^b)$ when $0 < a < b$.
- Any polynomial is asymptotically smaller than any exponential with base > 1 :

$$n^k = o(c^n).$$

- Exponential is asymptotically smaller than factorial:

$$c^n = o(n!).$$

Why This Matters in Algorithms

- It helps compare algorithms without machine-dependent constants.
 - It highlights which designs remain practical as data grows.
 - It explains why an $O(n \log n)$ algorithm is preferred over $O(n^2)$ for large inputs.
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Common Pitfalls

- Comparing runtimes only for small n and drawing asymptotic conclusions.
 - Forgetting that constants and lower-order terms can matter in practice for moderate input sizes.
 - Treating classes as equal when one is only an upper bound and not tight.
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Related

- [[Atomic/Big-O, Big-Omega, Big-Theta]]
- [[Lectures/Week 4]]